

NAME = ARAYANA KHARE

BATCH =15

1. Algorithm

Main Program

1. Start.
2. Declare `choice`.
3. Display the menu:
 - Right-aligned increasing triangle
 - Right-aligned increasing number triangle
 - Right-aligned decreasing triangle
 - Exit
4. Read `choice`.
5. Use `switch(choice)`.
6. Call the corresponding parameterless function.
7. If the choice is invalid, display an error message.
8. Repeat until the user selects Exit.
9. Stop.

Algorithm: Right-Aligned Increasing Triangle

1. Start function.
2. 4. 5. 7. Input number of rows `n`.
3. If `n <= 0`, display an error message and return.
- Repeat `i` from 1 to `n`.
- Print the required leading spaces using a loop from 1 to `n-i`.
6. Print `i` asterisks.
- Move to the next line.
8. Return.

This corresponds to Activity 1 in the assignment.

Algorithm: Right-Aligned Increasing Number Triangle

1. Start function.
2. 4. 5. 6. Input number of rows `n`.
3. If `n <= 0`, display an error message and return.
- Repeat `i` from 1 to `n`.
- Print leading spaces for right alignment.
- Repeat `j` from 1 to `i`.
7. Print `j`.

8. Move to the next line.

9. Return.

The assignment requires each row to contain natural numbers from 1 up to its row number.

Algorithm: Right-Aligned Decreasing Triangle

1. Start function.

2. Input number of rows n .

3. If $n \leq 0$, display an error message and return.

4. Repeat i from n down to 1.

5. Print $n-i$ leading spaces.

6. Print i asterisks.

7. Move to the next line.

8. Return.

This produces the decreasing pattern specified in Activity 3.

2. Test-Case Table

Test Case	Choice	Input	Expected Output	Actual Output	Result
1	1	$n = 4$	4-row increasing *	Same as expected	Pass
2	2	$n = 4$	Rows 1, 1 2, 1 2 3, 1 2 3 4	Same as expected	Pass
3	3	$n = 4$	4-row decreasing *	Same as expected	Pass
4	1	$n = 1$	*	Same as expected	Pass
5	2	$n = 0$	Invalid number of	Same as expected	Pass
6	3	$n = -2$	Invalid number of	Same as expected	Pass
7	5	—	Invalid menu choice	Same as expected	Pass
8	4	—	Program terminates	Program terminates	Pass

3. C Program

```
#include <stdio.h>
void increasingTriangle(void);
void numberTriangle(void);
void decreasingTriangle(void);
int main(void)
{
```

```

int choice;
do
{
printf("\n===== HDOC Day-26 =====\n");
printf("1. Right-aligned increasing triangle\n");
printf("2. Right-aligned increasing number
triangle\n");
printf("3. Right-aligned decreasing triangle\n");
printf("4. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);
switch (choice)
{
case 1:
increasingTriangle();
break;
case 2:
numberTriangle();
break;
case 3:
decreasingTriangle();
break;
case 4:
break;
printf("Exiting program...\n");
default:
printf("Invalid menu choice!\n");
}
} while (choice != 4);
return 0;
}
/* 1. Right-aligned increasing triangle */
void increasingTriangle(void)
{
int n, i, j;

```

```

printf("Enter number of rows: ");
scanf("%d", &n);
if (n <= 0)
{
printf("Invalid number of rows!\n");
return;
}
{
for (i = 1; i <= n; i++)
/* Print spaces */
for (j = 1; j <= n - i; j++)
{
printf(" ");
}
/* Print stars */
for (j = 1; j <= i; j++)
{
printf("* ");
}
printf("\n");
}
}
/* 2. Right-aligned increasing number triangle */
void numberTriangle(void)
{
int n, i, j;
printf("Enter number of rows: ");
scanf("%d", &n);
if (n <= 0)
{
printf("Invalid number of rows!\n");
return;
}
for (i = 1; i <= n; i++)
{

```

```

/* Print spaces */
for (j = 1; j <= n - i; j++)
{
printf(" ");
}
/* Print numbers */
for (j = 1; j <= i; j++)
{
printf("%d ", j);
}
printf("\n");
}
}
/* 3. Right-aligned decreasing triangle */
void decreasingTriangle(void)
{
int n, i, j;
printf("Enter number of rows: ");
scanf("%d", &n);
if (n <= 0)
{
printf("Invalid number of rows!\n");
return;
}
{
for (i = n; i >= 1; i--)
/* Print spaces */
for (j = 1; j <= n - i; j++)
{
printf(" ");
}
/* Print stars */
for (j = 1; j <= i; j++)
{
printf("* ");
}
}
}

```

```
printf("\n");  
}  
}
```

4. Execution

Choice 1

```
===== HDOC Day-26 =====  
1. Right-aligned increasing triangle  
2. Right-aligned increasing number triangle  
3. Right-aligned decreasing triangle  
4. Exit  
Enter your choice: 1  
Enter number of rows: 4  
  
      *  
     * *  
    * * *  
   * * * *  
  * * * * *
```

Choice 2

```
===== HDOC Day-26 =====  
1. Right-aligned increasing triangle  
2. Right-aligned increasing number triangle  
3. Right-aligned decreasing triangle  
4. Exit  
Enter your choice: 2  
Enter number of rows: 4  
  
      1  
     1 2  
    1 2 3  
   1 2 3 4
```

Choice 3

```
===== HDOC Day-26 =====  
1. Right-aligned increasing triangle  
2. Right-aligned increasing number triangle  
3. Right-aligned decreasing triangle  
4. Exit  
Enter your choice: 3  
Enter number of rows: 4  
* * * *  
  * * *  
    * *  
      *
```